

CHAPTER - 4

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION

The ONE8 Shoes E-Commerce Database System is designed to efficiently manage customers, products, categories, brands, shopping cart, wishlist, orders, payments, delivery, inventory, reviews, and promotional offers. Functional dependencies help identify the relationship between attributes in each table, while redundancy analysis and data anomaly detection ensure data consistency and reduce duplication. A well-designed database improves performance, maintains data integrity, and supports efficient management of the online shoe store.

1. CUSTOMER TABLE

Attribute	Description
Customer_ID	Unique customer ID
Customer_Name	Customer name
Email	Email address
Phone_Number	Contact number
Address	Shipping address
Password	Login password

Functional Dependency

Customer_ID → Customer_Name, Email, Phone_Number, Address, Password

Redundancy Analysis

Customer information is stored only once in the Customer table. Orders, the shopping cart, wishlist, and reviews store only the Customer_ID, preventing duplicate customer records.

Data Anomaly:

No data anomalies are present because customer information is stored only once in the Customer table. This prevents insert, update, and delete anomalies and ensures data consistency.

2. ADMIN TABLE

Attribute	Description
Admin_ID	Unique admin ID
Admin_Name	Administrator name
Email	Email address
Password	Login password

Functional Dependency

Admin_ID → Admin_Name, Email, Password

Redundancy Analysis

Administrator information is stored separately from the Customer table to avoid repeating login details across different tables.

Data Anomaly:

No data anomalies are present because administrator information is stored only once in the Admin table. This eliminates duplicate records and ensures easy maintenance of administrator details.

3. CATEGORY TABLE

Attribute	Description
Category_ID	Unique category ID
Category_Name	Category name

Functional Dependency

Category_ID → Category_Name

Redundancy Analysis

Each category is stored only once. Products reference the category using Category_ID instead of repeating the category name.

Data Anomaly:

No data anomalies are present because category details are maintained in a separate table. This avoids duplication and ensures that updates to category information are made only once.

4. BRAND TABLE

Attribute	Description
Brand_ID	Unique brand ID
Brand_Name	Brand name

Functional Dependency

Brand_ID → Brand_Name

Redundancy Analysis

Each brand is stored only once. Products reference the brand using Brand_ID instead of repeating the brand name.

Data Anomaly:

No data anomalies are present because brand details are maintained in a separate table, avoiding duplication and keeping brand updates consistent across all related products.

5. PRODUCT TABLE

Attribute	Description
Product_ID	Unique product ID
Product_Name	Product name
Price	Product price
Stock	Available stock
Category_ID	Category ID
Brand_ID	Brand ID

Functional Dependency

Product_ID → Product_Name, Price, Stock, Category_ID, Brand_ID

Redundancy Analysis

Product details are stored only in the Product table. The cart, wishlist, order items, and reviews use Product_ID instead of repeating product information.

Data Anomaly:

No data anomalies are present because product information is stored independently and linked to categories and brands using foreign keys. This prevents duplicate product records and maintains data integrity.

6. SHOPPING_CART TABLE

Attribute	Description
Cart_ID	Unique cart entry ID
Customer_ID	Customer ID
Product_ID	Product ID
Quantity	Quantity selected

Functional Dependency

Cart_ID → Customer_ID, Product_ID, Quantity

Redundancy Analysis

The Shopping_Cart table connects customers and products without repeating customer or product details, which remain stored in their own tables.

Data Anomaly:

No data anomalies are present because the Shopping_Cart table stores only the relationship between a customer and a product. This avoids duplicate customer or product information.

7. WISHLIST TABLE

Attribute	Description
Wishlist_ID	Unique wishlist entry ID
Customer_ID	Customer ID
Product_ID	Product ID

Functional Dependency

Wishlist_ID → Customer_ID, Product_ID

Redundancy Analysis

The Wishlist table stores only the link between a customer and a saved product, avoiding repetition of customer or product details.

Data Anomaly:

No data anomalies are present because the Wishlist table stores only relationship data. Customer and product details are maintained independently in their own tables.

8. ORDERS TABLE

Attribute	Description
Order_ID	Unique order ID
Customer_ID	Customer ID
Order_Date	Date of order
Total_Amount	Total order amount
Order_Status	Order status

Functional Dependency

Order_ID → Customer_ID, Order_Date, Total_Amount, Order_Status

Redundancy Analysis

The Orders table stores only order-related information. Customer and product details are referenced using foreign keys instead of being repeated.

Data Anomaly:

No data anomalies are present because order details are stored separately and linked to customers through foreign keys. This design prevents data inconsistency and unnecessary duplication.

9. ORDER_ITEM TABLE

Attribute	Description
Order_Item_ID	Unique order item ID
Order_ID	Order ID
Product_ID	Product ID
Quantity	Quantity ordered
Price	Product price at time of order

Functional Dependency

Order_Item_ID → Order_ID, Product_ID, Quantity, Price

Redundancy Analysis

This table connects orders and products, avoiding repeated product information in the Orders table.

Data Anomaly:

No data anomalies are present because the Order_Item table stores only the relationship between orders and products. This avoids duplicate product information and supports accurate order management.

10. PAYMENT TABLE

Attribute	Description
Payment_ID	Unique payment ID
Order_ID	Order ID
Payment_Method	Payment method
Payment_Status	Payment status

Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Status

Redundancy Analysis

Payment information is stored separately from the Orders table to avoid duplicate payment records.

Data Anomaly:

No data anomalies are present because payment details are maintained in a separate table. This ensures that payment information can be updated without affecting order records.

11. DELIVERY TABLE

Attribute	Description
Delivery_ID	Unique delivery ID
Order_ID	Order ID
Tracking_Number	Shipment tracking number
Delivery_Status	Delivery status

Functional Dependency

Delivery_ID → Order_ID, Tracking_Number, Delivery_Status

Redundancy Analysis

Delivery details are maintained in a separate table, preventing repeated shipment information from being stored inside the Orders table.

Data Anomaly:

No data anomalies are present because delivery details are stored independently from order information. This prevents data duplication and allows delivery records to be managed efficiently.

12. INVENTORY TABLE

Attribute	Description
Inventory_ID	Unique inventory record ID
Product_ID	Product ID
Stock_Quantity	Current stock quantity
Reorder_Level	Minimum stock threshold

Functional Dependency

Inventory_ID → Product_ID, Stock_Quantity, Reorder_Level

Redundancy Analysis

Stock information is maintained only in the Inventory table and referenced using Product_ID, instead of being repeated inside the Product table.

Data Anomaly:

No data anomalies are present because inventory details are stored separately from product details. This keeps stock updates isolated and consistent across the system.

13. REVIEWS TABLE

Attribute	Description
Review_ID	Unique review ID
Customer_ID	Customer ID
Product_ID	Product ID
Rating	Rating value
Comment	Review comment

Functional Dependency

Review_ID → Customer_ID, Product_ID, Rating, Comment

Redundancy Analysis

The Reviews table stores only the relationship between a customer and a product, along with the review content, avoiding repetition of customer or product details.

Data Anomaly:

No data anomalies are present because reviews are stored independently and linked to customers and products using foreign keys, preventing duplicate customer or product data.

14. OFFERS / COUPON TABLE

Attribute	Description
Coupon_ID	Unique coupon ID
Coupon_Code	Coupon code
Discount_Value	Discount amount or percentage
Valid_From	Validity start date
Valid_To	Validity end date

Functional Dependency

Coupon_ID → Coupon_Code, Discount_Value, Valid_From, Valid_To

Redundancy Analysis

Coupon details are stored only once in the Offers table. Orders reference an applied coupon using Coupon_ID instead of repeating discount details.

Data Anomaly:

No data anomalies are present because coupon information is maintained independently in its own table, keeping promotional data consistent and easy to update.

CONCLUSION

The ONE8 Shoes E-Commerce Database System is designed using proper functional dependencies and normalized tables. Each entity stores only its relevant data, reducing redundancy and preventing insert, update, and delete anomalies. This design improves data consistency, maintains integrity, and ensures efficient database management for the online shoe store.